

Basic Principles of Medicine 1
Module : Foundations to Medicine
Lecture No: 41

**Title: Purine Nucleotide Synthesis and
Salvage pathway**

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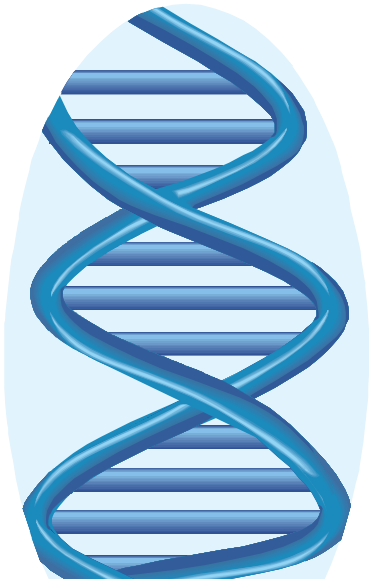
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Recommended Reading

- **Chapter 22: Nucleotide metabolism (Lippincott's Biochemistry 9th edition)**
 - <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
 - [Nucleotide Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - Lippincott's questions: page 306, 22.1-22.5
- **Refer: Focused: Case 7: Joint pain**
 - [Clinical Cases | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

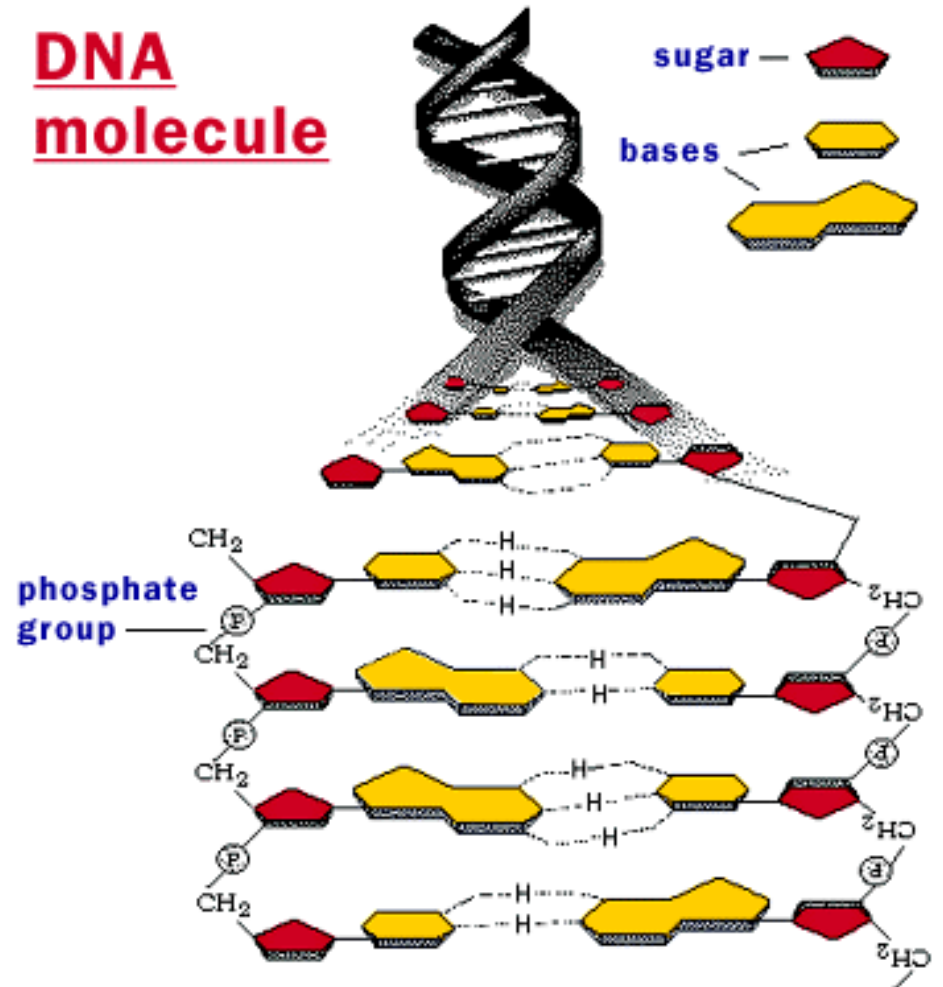
Purine Nucleotide Synthesis and Salvage Pathway



Reference: Chapter 22 (Lippincott's Biochemistry)

Refer: Case 7: Joint pain page 510;

Lippincott's questions: page 306, 22.1-22.5



Objectives: Purine nucleotide synthesis and salvage pathway

SOM.MK.I.BPM1.1.FTM.3.BCHM.0020	Distinguish between purines, pyrimidines; ribonucleosides, deoxyribonucleosides; ribonucleotides and deoxyribonucleotides.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0162	Enumerate C and N donors of purine ring (amino acids and 1-C groups)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0163	Outline denovo purine biosynthesis and indicate the role of pentose phosphate pathway as donors of pentoses (PRPP). Review formation of AMP and GMP from IMP
SOM.MK.I.BPM1.1.FTM.3.BCHM.0164	Discuss the regulation of denovo purine biosynthesis (Identify the regulatory enzyme and positive and negative modulators)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0165	Correlate mechanism of action of sulfa drugs, trimethoprim, methotrexate, mycophenolic acid to their clinical application
SOM.MK.I.BPM1.1.FTM.3.BCHM.0166	Explain formation of deoxyribonucleotides from ribonucleotides (Specify coenzyme and inhibitor)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0167	Identify reactions catalyzed by HGPRT and APRT
SOM.MK.I.BPM1.1.FTM.3.BCHM.0168	Correlate clinical, laboratory (hyperuricemia) and molecular findings in Lesch-Nyhan syndrome (Specify enzyme deficient).
SOM.MK.III.BPM1.1.FTM.3.BCHM.0084	Describe the role of thiopurine methyl transferase (TPMT) in the conversion of 6-mercaptopurine to an inactive product.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0085	Explain the mechanism of 6-mercaptopurine/azathioprine toxicity in patients who are homozygous for autosomal recessive polymorphisms in the TPMT gene (example of an adverse drug reaction).

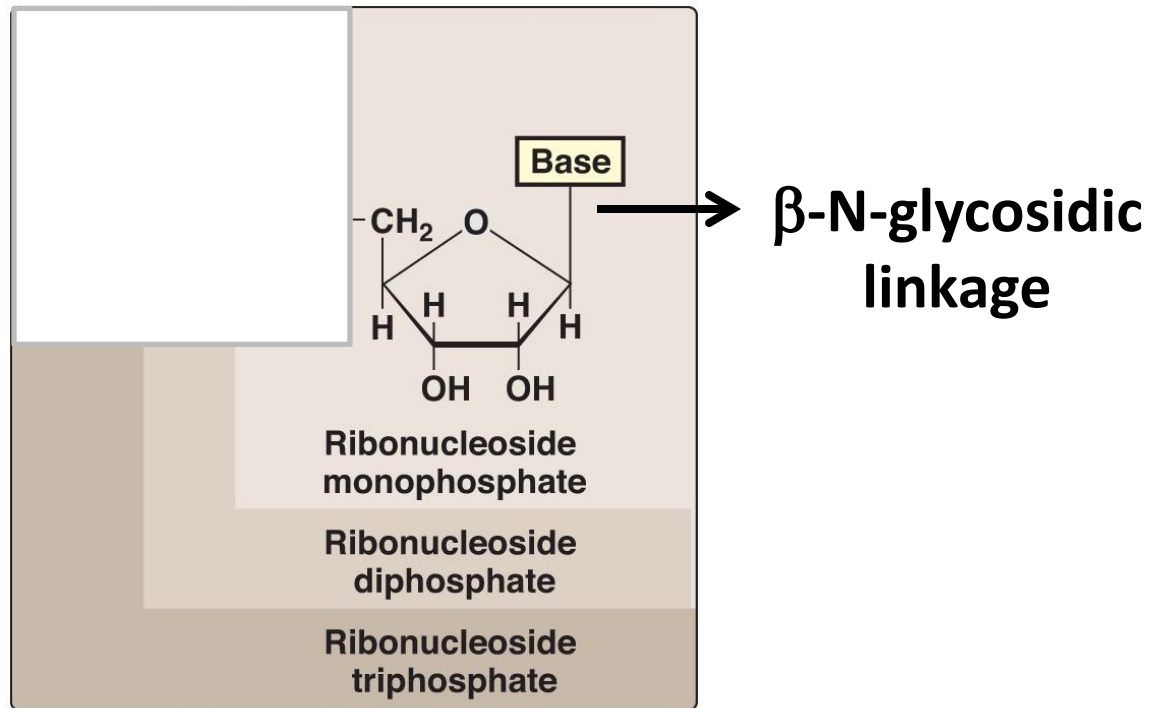
Tips for covering the biochemistry and genetic components of medical school

- Genetics: **Definitions**, population genetics
- Biochemistry: **Pathways, Regulation of pathways** and molecular structures.
- **Clinical and pharmacological correlations**

Overview of functions of nucleotides

- Ribonucleotides (contain ribose) building blocks of RNA
- Deoxyribonucleotides (contain deoxyribose) building blocks of DNA
- ATP is 'energy currency' of cells
- Components of coenzymes like FAD, NAD, NADP and coenzyme A
- Nucleotides like cAMP and cGMP are intracellular second messengers
- Inhibitors of purine and pyrimidine nucleotide synthesis are used as **anti-cancer drugs or immunosuppressive drugs (rapidly dividing cells) or antimicrobials**

Review of nucleoside and nucleotide



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- N-containing base + Ribose = Nucleoside
- **β -N-glycosidic linkage** between base and ribose
- N-containing base + Ribose + Phosphate = Nucleotide
- Purine base + Ribose = Purine nucleoside
- Pyrimidine base + Ribose = Pyrimidine nucleoside

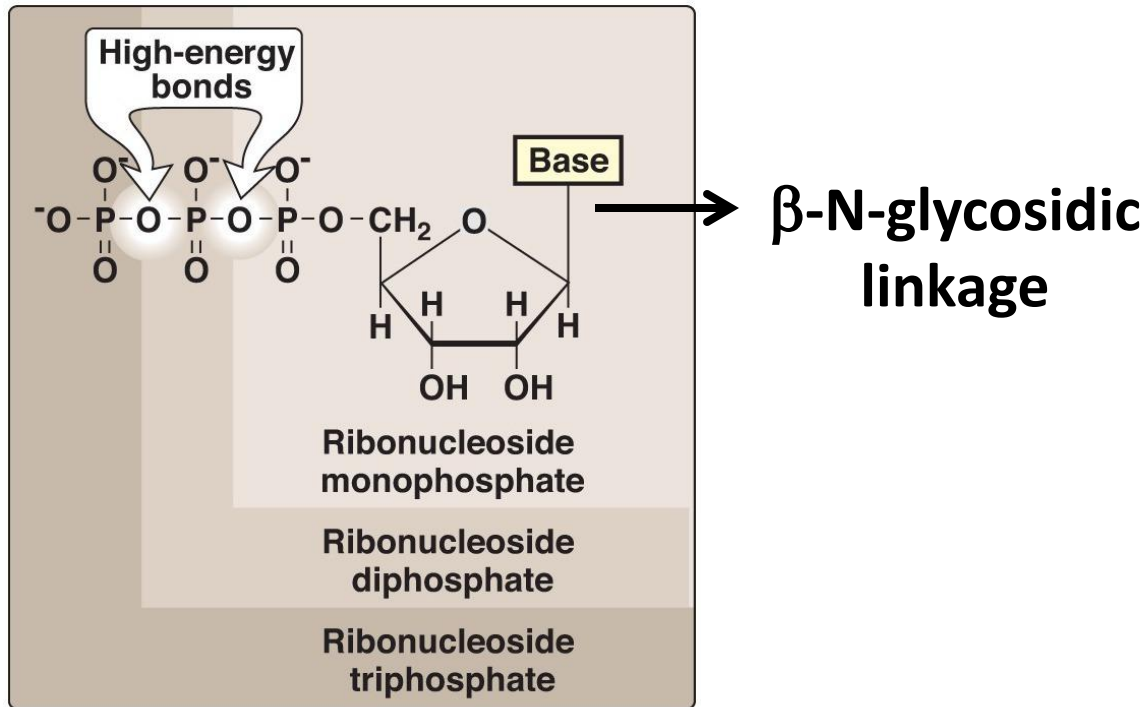
DNA and RNA Purines		
Adenine (A)	Guanine (G)	
RNA Pyrimidines		
Thymine (T)	Cytosine (C)	Uracil (U)
DNA Pyrimidines		

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Mnemonic: Pur-AG TC
English: Pure Silver Taxi



Review of nucleoside and nucleotide



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DNA and RNA Purines		
Adenine (A)	Guanine (G)	
RNA Pyrimidines		
Thymine (T)	Cytosine (C)	Uracil (U)
DNA Pyrimidines		

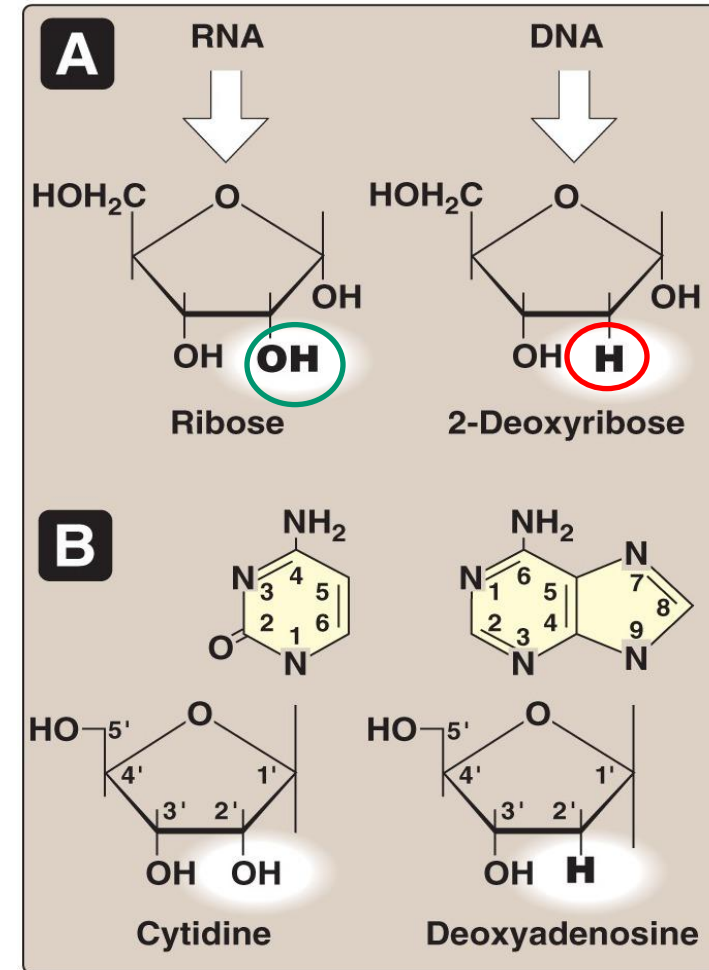
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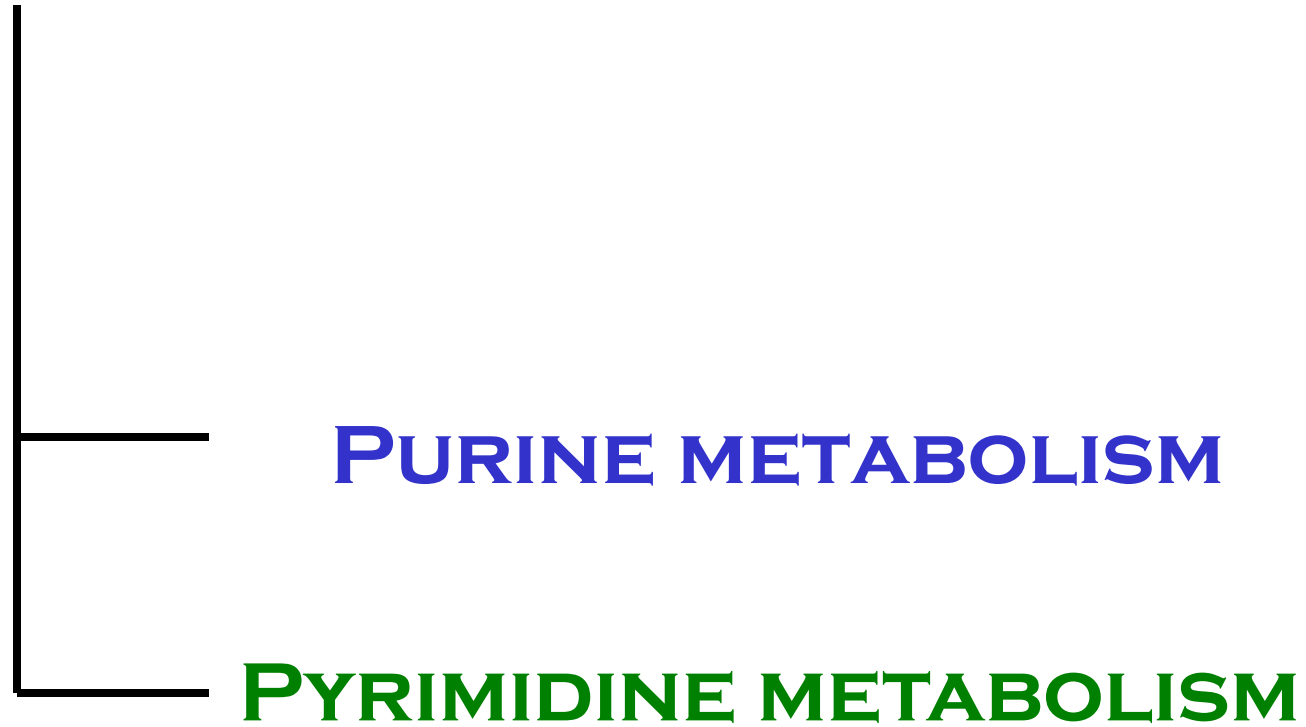
Review of ribonucleotides vs deoxyribonucleotides

- Base + **Ribose** + Phosphate = Ribonucleotide. Building blocks of RNA
- Base + **Deoxyribose** + Phosphate = Deoxyribonucleotide. Building blocks of DNA
- Ribose sugar and phosphate donor **PRPP** in purine and pyrimidine nucleotide biosynthesis
- Purine and pyrimidine biosynthesis inhibitors: **Anticancer or antimicrobial agents**



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Nucleotide metabolism



Overview of purine nucleotide metabolism

Amino acids (Glycine, aspartate, glutamine) and Formyl tetrahydrofolate

Purine
nucleotide
denovo synthesis

Purine nucleotides
(IMP, AMP and GMP)

ATP and GTP

dATP and dGTP

DNA

(During replication)

RNA

(Transcription)

Nucleic acids

Uric acid

(End product of Purine base degradation)

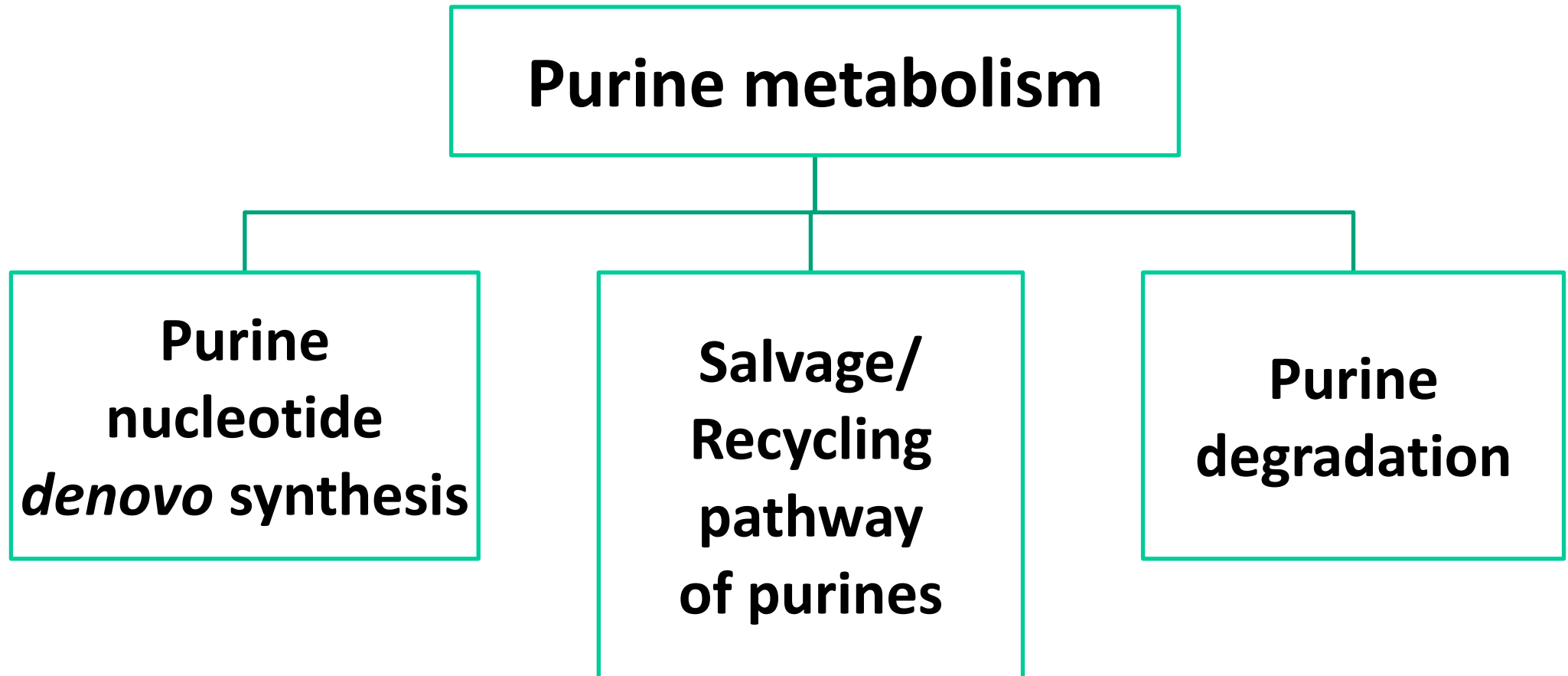
Salvage/ Recycling
pathway
of purines

Adenine, hypoxanthine and
Guanine (Purine bases)

Nucleic acid turnover/
cell death

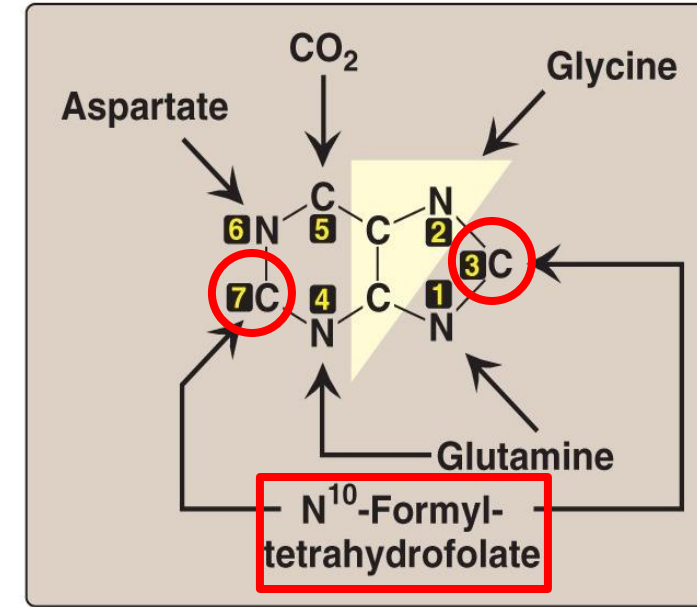
Purine
degradation

Purine nucleotide metabolism



Purine nucleotide *denovo* synthesis

- Precursors
 - Amino acids (glutamine, aspartate, glycine)
 - Formyl tetrahydrofolate (THF) (**folic acid/ vitamin B9**)
- Each donor donates specific C or N atom
- C and N atoms added to Ribose phosphate (Numbers on purine in diagram is order in which added)
- Energy (~6ATP/ purine synthesis)
- Ribose phosphate from **Pentose phosphate pathway (HMP shunt)**
- Remember: 2 C-atoms of purine from formyl tetrahydrofolate (**folic acid/ vitamin B9**)

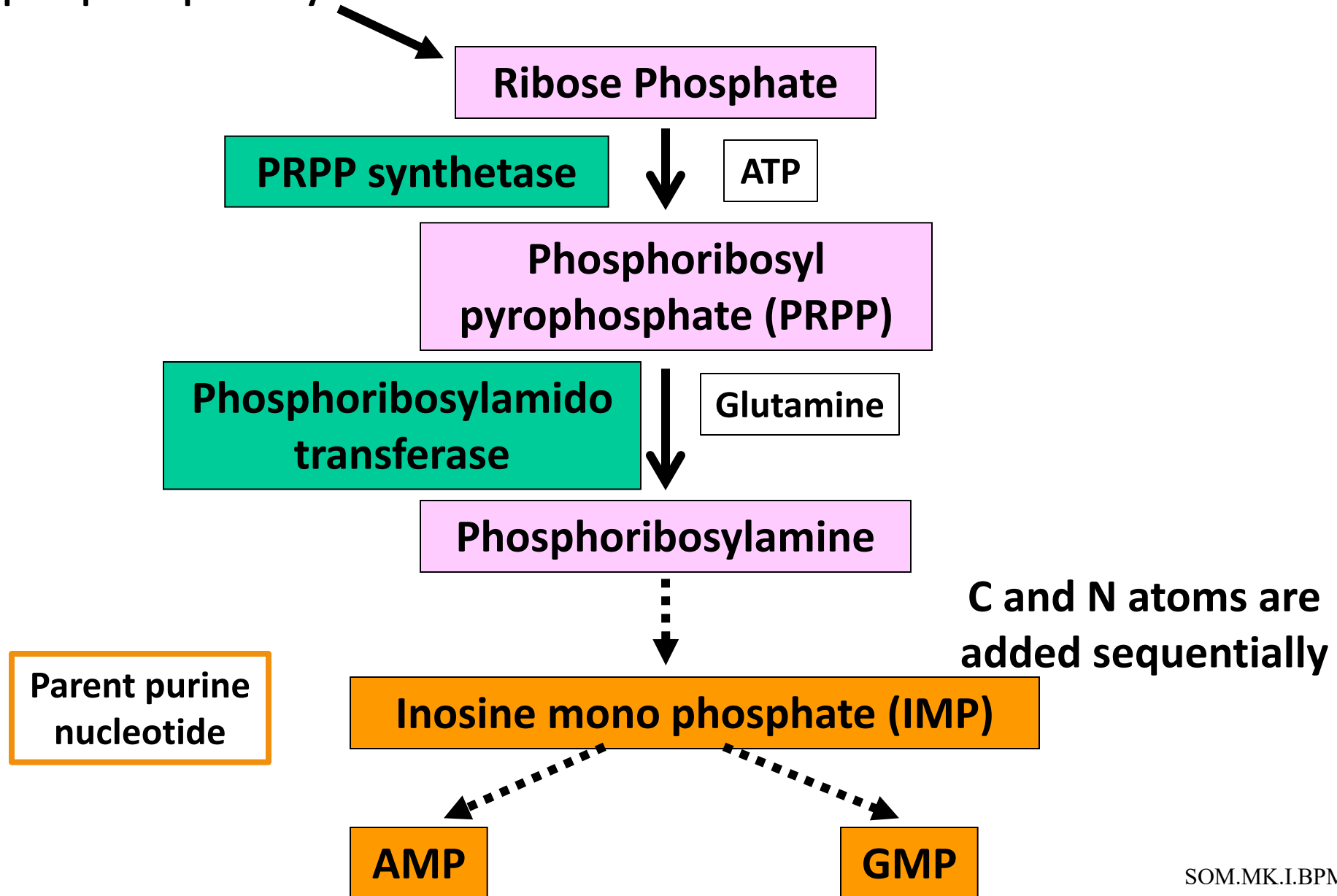


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**Donors of C and N
atoms of the purine ring**

Denovo purine nucleotide biosynthesis

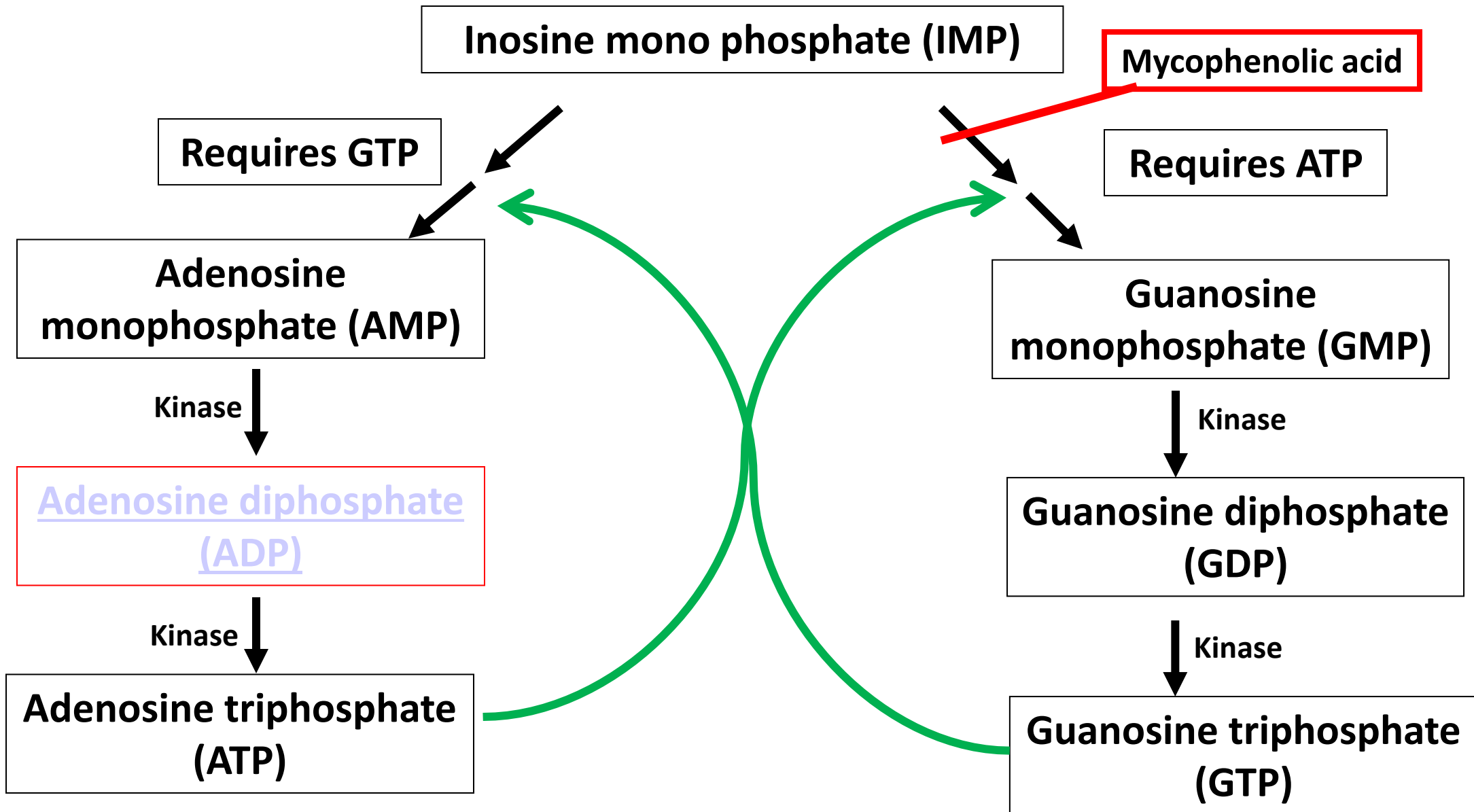
Pentose phosphate pathway



Purine nucleotide synthesis: IMP Formation

- **PRPP synthetase:** Regulated enzyme
- **Phosphoribosylamidotransferase:** Committed step (Regulated step) – N donated by glutamine
- Inosine monophosphate (**IMP**) (Hypoxanthine-ribose-phosphate): Parent purine nucleotide
 - Forms AMP and GMP (Common precursor)

Conversion of IMP to AMP and GMP



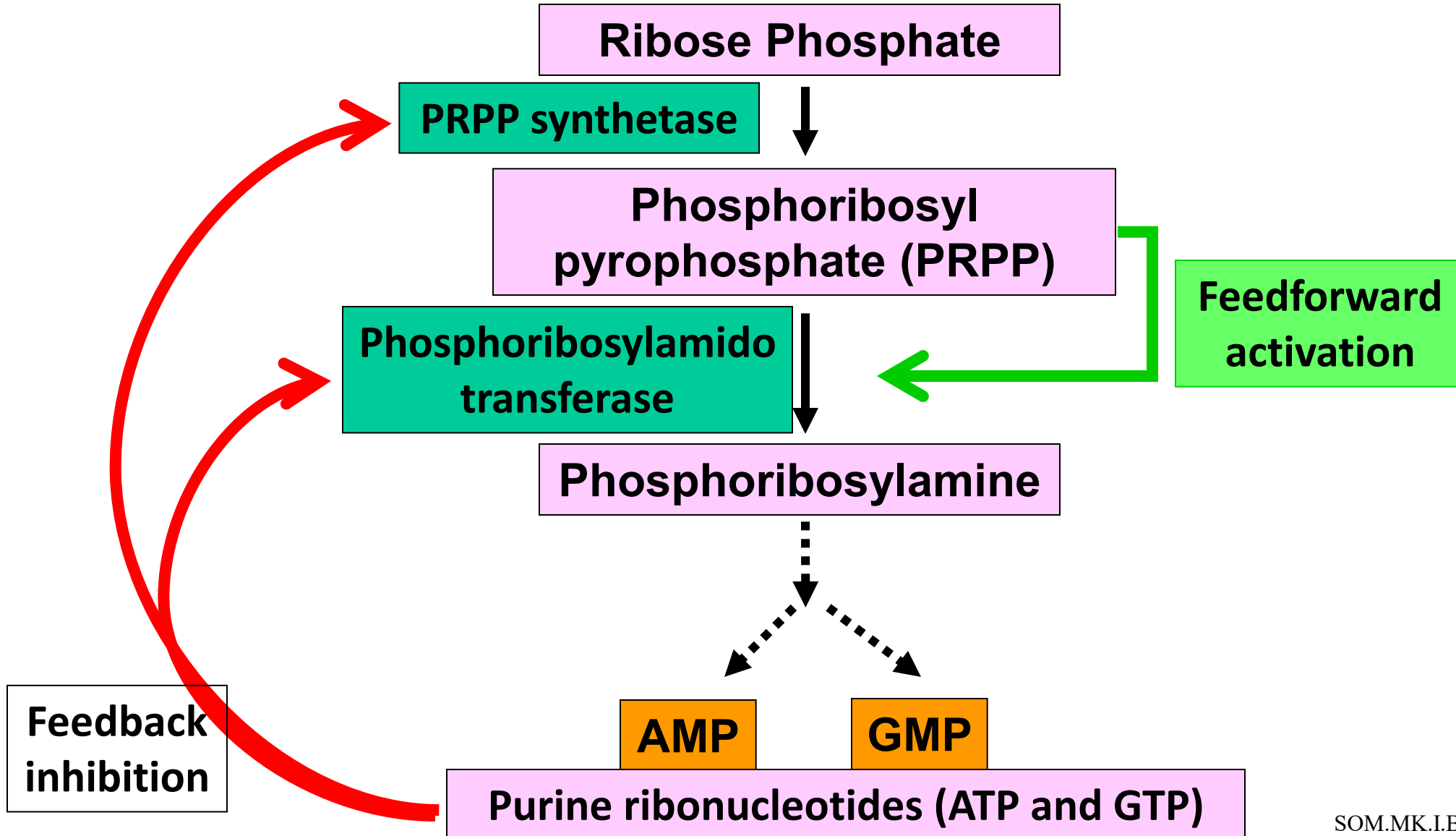
Conversion of IMP to AMP and GMP

- Formation of AMP from IMP requires GTP
- Formation of GMP from IMP requires ATP
- Ensures cellular $[ATP] = [GTP]$
- **Mycophenolic acid** inhibits conversion of IMP to GMP
 - Inhibits **IMP dehydrogenase**
 - Used following transplantation to prevent rejection
- **Ribonucleotide reductase** forms deoxyribonucleotides (dADP/dGDP)

Regulation of purine biosynthesis

- **Regulated enzymes:**
 - PRPP synthetase
 - Phosphoribosylamidotransferase
- PRPP also used for pyrimidine biosynthesis and salvage pathway
- Feedback/product inhibition: Purine nucleotides (ATP and GTP) inhibit PRPP synthetase and the phosphoribosylamidotransferase
- **PRPP: Feedforward activator of Phosphoribosylamidotransferase.**
 - High [PRPP] increases purine nucleotide synthesis

Regulation of de novo purine biosynthesis



Folate metabolism (vitamin B9) inhibitors (bacteria and humans)

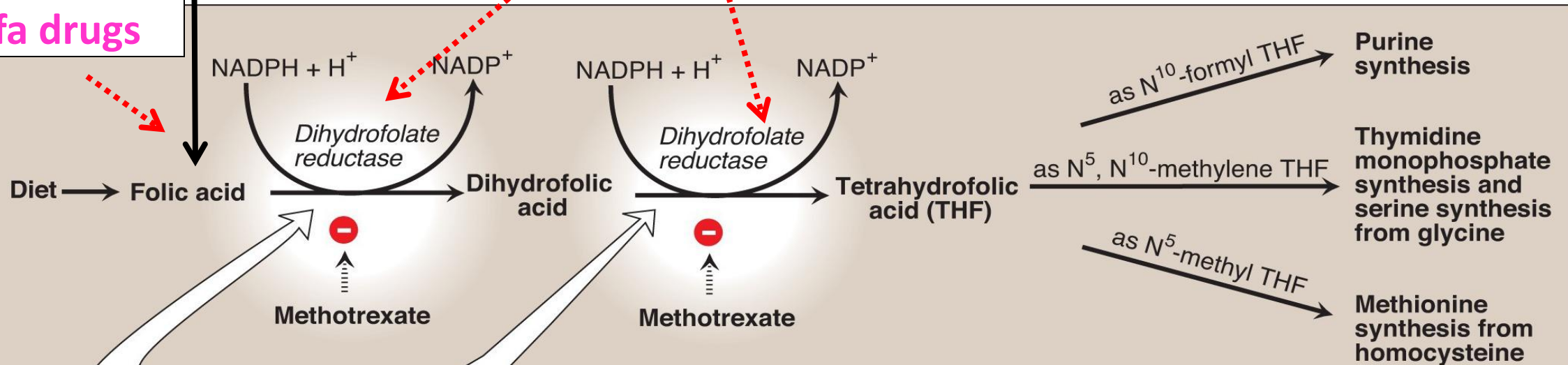
Bacteria

Para amino benzoic
acid (PABA)

Trimethoprim

2 C-atoms from formyl-tetrahydrofolate

Sulfonamide/
Sulfa drugs



Dihydrofolate reductase is competitively inhibited by methotrexate, a folic acid analogue used to treat psoriasis, rheumatoid arthritis, and neoplastic diseases.

Inhibitors of **bacterial** folate metabolism

- In microorganisms, folic acid synthesized from PABA (p-amino benzoic acid)
- PABA analogs (**sulfonamides; sulfa drugs**) **inhibit bacterial folate synthesis**
- **Trimethoprim** inhibits bacterial **dihydrofolate reductase**
- Sulfonamide- trimethoprim: Antibacterial agent
- Reduce tetrahydrofolate and purine nucleotide synthesis
- Humans cannot synthesize folic acid (Folic acid obtained in diet), sulfonamide- trimethoprim combination does not affect purine synthesis in humans

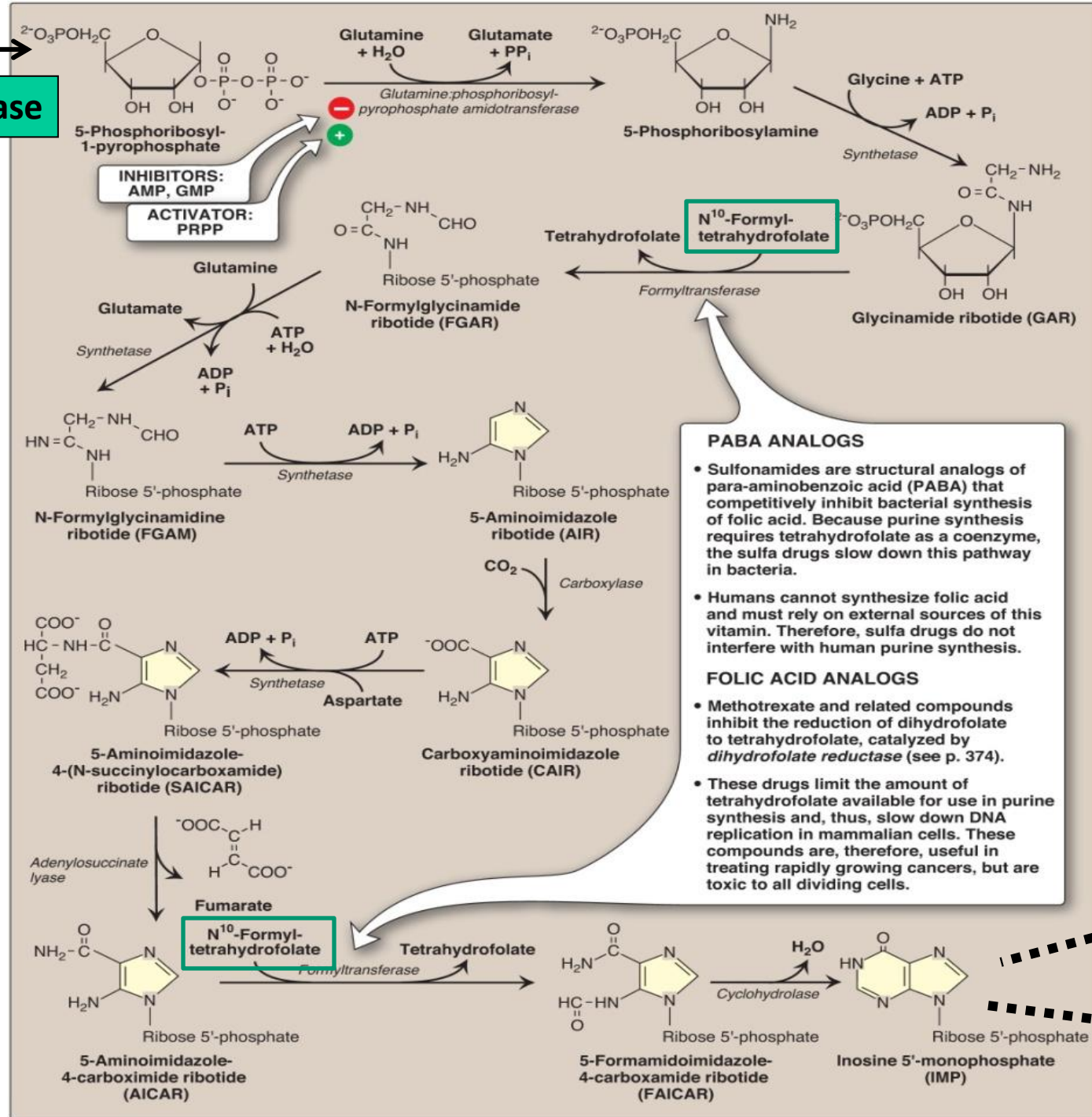
Inhibitors of **eukaryotic** purine nucleotide synthesis

- Humans obtain folate (vitamin B9) from diet
- **Methotrexate** (folate analog): **Anticancer agent**
- Competitive inhibitor of **dihydrofolate reductase**
- Reduces tetrahydrofolate for purine synthesis and slows down DNA replication (cell division) in **mammalian cells**
- Side effects on all rapidly dividing cells: anemia, GI disturbance, scaly skin, hair loss, immune deficiency

Ribose Phosphate

PRPP synthetase

Summary of purine nucleotide synthesis



GTP

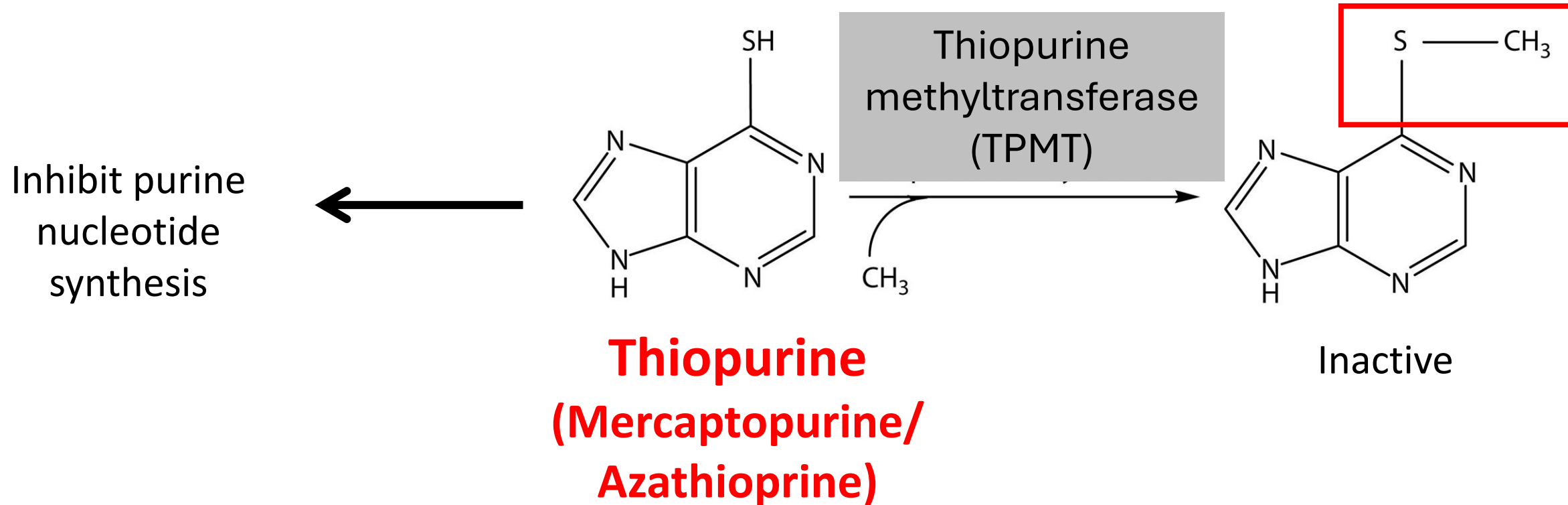
ATP

Thiopurine: Mercaptopurine and azathioprine

- Thiopurine: Mercaptopurine and azathioprine inhibit de novo purine synthesis (inhibit cell division)
 - Used as Anticancer agents (leukemia) and immunosuppressive agents
- Thiopurine → Inactive metabolites by thiopurine methyltransferase (TPMT)
- TPMT variants → Low activity of TPMT → Less inactivation of thiopurine → Toxic effects of thiopurine (Bone marrow suppression/ low neutrophil count/ aplastic anemia)
- **Pharmacogenetics:** Test for TPMT variants before administration
 - **Lower drug dose if TPMT variants present**



Thiopurine methyltransferase (TPMT)



- Thiopurine → Inactive metabolites by thiopurine methyl transferase (TPMT)
- TPMT variants → Low activity of TPMT → Less inactivation of thiopurine → Toxic effects of thiopurine (Bone marrow suppression/ low neutrophil count/ aplastic anemia)

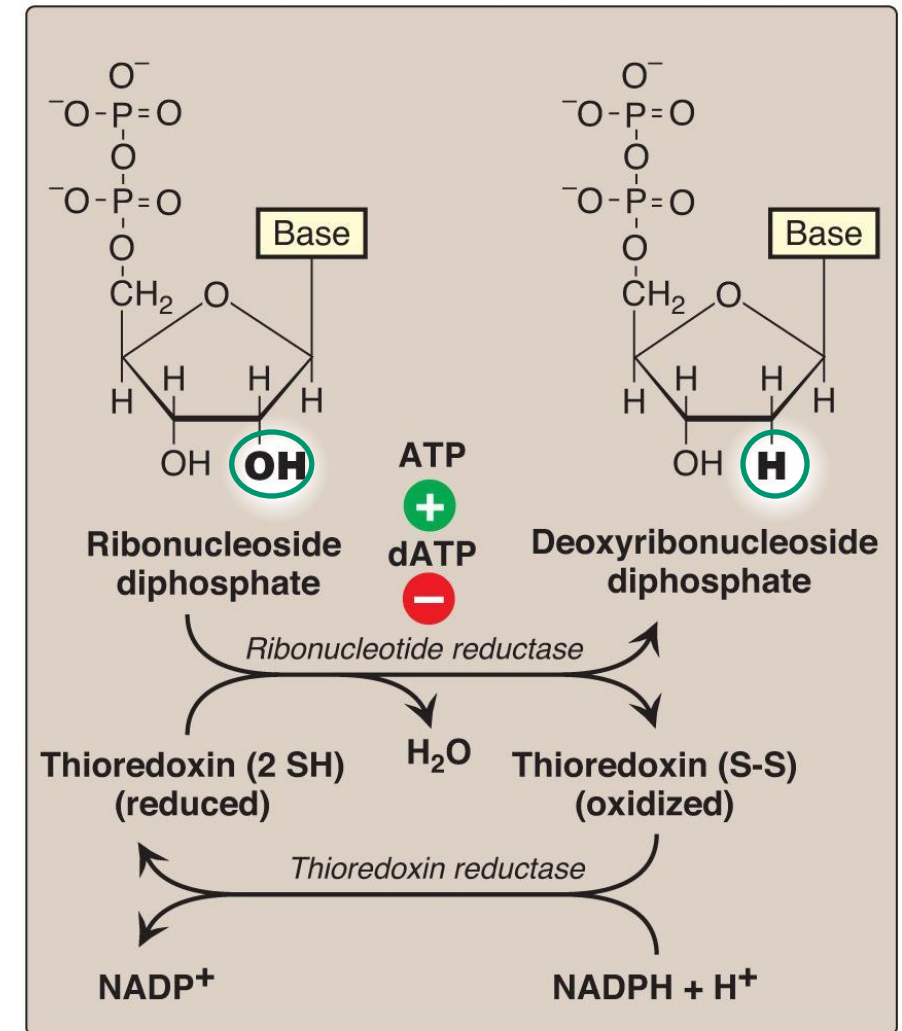
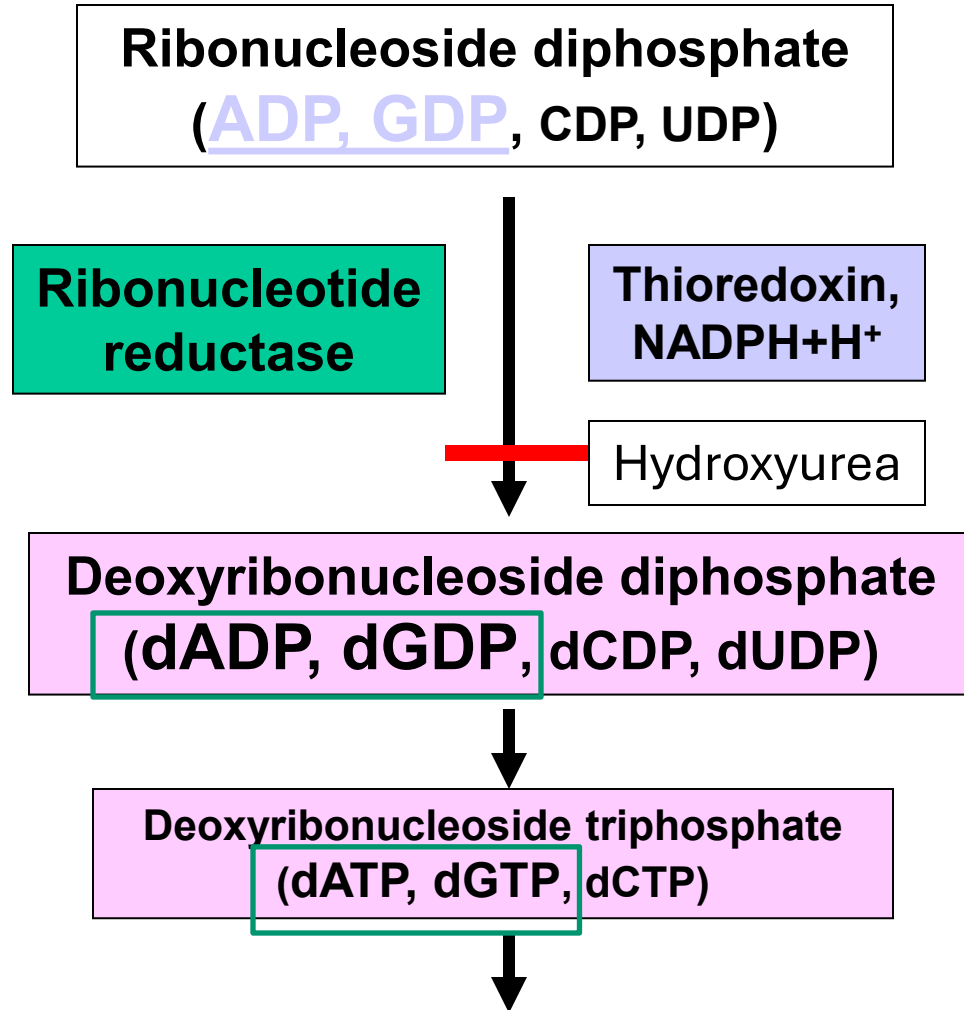


Folic acid (folate/ vitamin B9): Requirement and deficiency

- Formyl-THF or methylene-THF used for purine nucleotide and thymidine synthesis (dTMP synthesis)
- Essential for DNA synthesis during S phase of cell cycle
- Critical for rapidly dividing cells
- Folate deficiency: **Macrocytic anemia** and decreased cell division; Impaired purine and pyrimidine nucleotide synthesis
- Pre and perinatal folate deficiency in pregnancy
 - Higher risk of fetal **neural tube defects**



Conversion of ribonucleotides to deoxyribonucleotides



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DNA Replication (S phase) during cell division

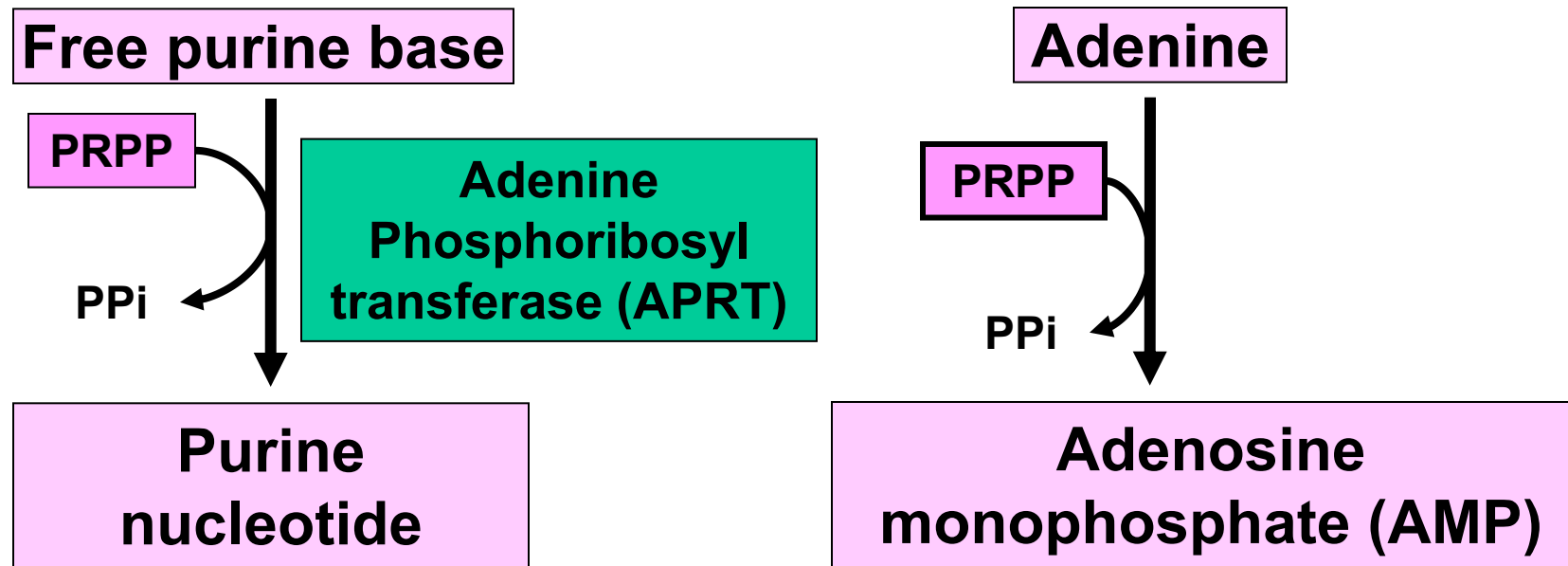
Conversion of ribonucleotides to deoxyribonucleotides

- **Ribonucleotide reductase**
- Converts **purine and pyrimidine ribonucleotides** to deoxyribonucleotides
- Ribose sugar converted to deoxyribose
- Requires: Thioredoxin (protein coenzyme) and NADPH+ H⁺
- Regulated to ensure equivalent concentrations of dATP, dGTP, dCTP and dUTP during S phase of cell cycle
- Inhibited by high **[dATP]** (refer to SCID)
- **Hydroxyurea** (anticancer agent): Inhibits ribonucleotide reductase

Salvage/ recycling pathway of purines

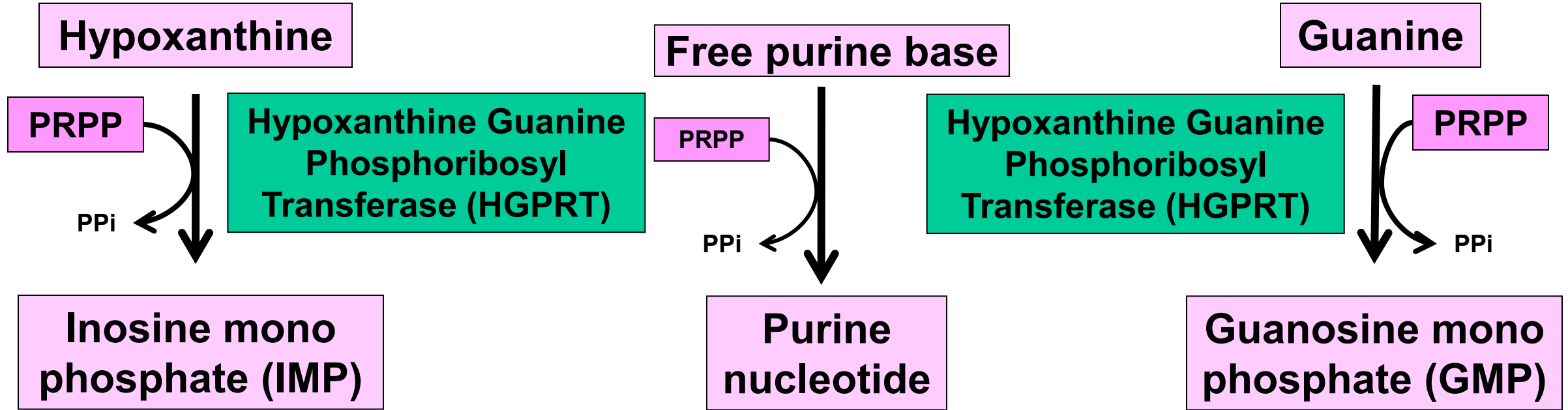
- Purine bases recycled (reused) by salvage pathway
- Free purine bases converted to **purine nucleotides**
 - Free bases derived from cellular DNA lysis (following cell death) and RNA (cell turnover)
- About **80%** of bases are reused
- Uses **less energy** than denovo synthesis pathway
- **HGPRT** and APRT are enzymes of purine salvage pathway
- Salvage pathway is important for **brain**

Salvage pathway: Adenine phosphoribosyl transferase (APRT)



- **PRPP** donates ribose-phosphate to form AMP

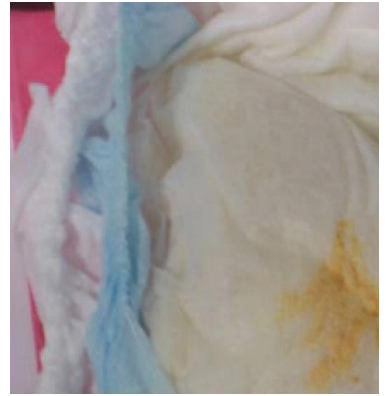
Salvage pathway: HGPRT



- Complete enzyme deficiency causes **Lesch-Nyhan syndrome**
- Purine bases NOT reused → Increased purine degradation → **Uric acid**
- High [PRPP] stimulates denovo synthesis of purine nucleotides
- **Hyperuricemia, Behavioral disturbances (Self mutilation)**
 - Remember, salvage pathway is important for brain

Lesch-Nyhan syndrome

- A 2-year-old **boy** evaluated for **developmental delay**, irritability, and failure to thrive. Examination shows truncal hypotonia, **spasticity and dystonia**.
- **X-linked recessive**
- Mother says **orange-colored crystals** in diapers
- **Serum uric acid is elevated** at 9.9 mg/dL
- Hypoxanthine-guanine-phosphoribosyltransferase (HGPRT) activity is absent
- Older children, **self-mutilation** (biting of lips and fingers), and **intellectual disability**
- Monosodium urate deposits in joints and soft tissue



Concept map p.305

